# Shengkai Zhang

*Curriculum Vitae*

## PERSONAL DETAILS


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|---|---|
| *Title* | Associate Professor |
| *Affiliation* | School of Navigation, Wuhan University of Technology |
| *Email* | shengkai@whut.edu.cn |
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## EDUCATION

**Ph.D. in Information and Communication Engineering** 09/2017 - 06/2021

*Huazhong University of Science and Technology (HUST)*

School of Electronic Information and Communications. Advisor: **Prof. Wei Wang**.

**M.Phil. in Computer Science and Engineering** 09/2012 - 11/2014

*Hong Kong University of Science and Technology (HKUST)*

Department of Computer Science and Engineering. Advisor: **Prof. Bo Li**.

**M.Sc. in Communication and Information Systems** 09/2009 - 06/2012

*Huazhong University of Science and Technology (HUST)*

School of Electronic Information and Communications. Advisor: **Prof. Hongbo Jiang**.

**B.Eng. in Communication Engineering** 09/2005 - 06/2009

*Central China Normal University (CCNU)*

## RESEARCH INTERESTS

Autonomous Maritime Navigation, Gaussian Splatting, SLAM, and Multisensor Fusion

## RESEARCH AND INDUSTRY EXPERIENCE

**Associate Professor** 03/2026 - Present

*School of Navigation, Wuhan University of Technology, China*

**Associate Professor** 06/2022 - 03/2026

*School of Information Engineering, Wuhan University of Technology, China*

**Assistant Professor** 06/2021 - 06/2022

*School of Information Engineering, Wuhan University of Technology, China*

**Research Assistant** 12/2014 - 06/2017

*HKUST-DJI Joint Innovation Laboratory, Hong Kong*

Interdisciplinary research on wireless sensing and autonomous navigation, together with resource-allocation algorithms for distributed computing systems.

## PUBLICATIONS


[1] **Shengkai Zhang**, Yuhe Liu, Jianhua He, Xuedou Xiao, Mozi Chen, and Kezhong Liu. *MOGS: Monocular Object-guided Gaussian Splatting in Large Scenes*. IEEE International Conference on Robotics and Automation (ICRA), 2026.

[2] Yiwen Zhou, Kezhong Liu, Xuedou Xiao, Xuming Zeng, **Shengkai Zhang**, and Mozi Chen. *Underwater 3D Gaussian Reconstruction Using Multi-view Sonar Imaging*. IEEE Sensors Letters, Early Access, 2026.

[3] Mengchuang Hu, Mozi Chen, Yiling Ren, Junjie Weng, Xuedou Xiao, **Shengkai Zhang**, and Kezhong Liu. *SIREN: Self-supervised 4D Implicit Representation for Enhanced Radar- and Vision-based Activity Recognition for Ship Watch-Keeping Officers*. IEEE Transactions on Mobile Computing, Early Access, 2026.

[4] Yiwen Zhou, Kezhong Liu, Mozi Chen, Xuming Zeng, Mengda Chen, Xuedou Xiao, and **Shengkai Zhang***. *Pose-Induced Error Compensation in Monocular Vision for Long-Range Air Draft Estimation*. IEEE Transactions on Intelligent Transportation Systems, Early Access, 2026. DOI: 10.1109/TITS.2026.3703866.

[5] Kezhong Liu, Cong Fan, **Shengkai Zhang***, Mozi Chen, Xuedou Xiao, Shuai Wang, Zheng Yang, and Wei Wang. *Robust mmWave Radar Sensing with Multi-sensor Temporal Calibration and Supervision*. IEEE Internet of Things Journal, Early Access, 2026. DOI: 10.1109/JIOT.2026.3668667.

[6] Kezhong Liu, Yiwen Zhou, Mozi Chen, Jianhua He, Jingao Xu, Zheng Yang, Chris Xiaoxuan Lu, and **Shengkai Zhang***. *mmWave Radar Perception Learning Using Pervasive Visual-Inertial Supervision*. IEEE Transactions on Intelligent Transportation Systems, Early Access, 2026. DOI: 10.1109/TITS.2026.3659202.

[7] Dashuai Pei, Yiwen Wu, Jianhua He, Kezhong Liu, Mozi Chen, Xuedou Xiao, **Shengkai Zhang**, and Jiawei Zheng. *Methodology and Benchmark for Automated Driving Theory Test of Large Language Models*. IEEE Transactions on Intelligent Transportation Systems, vol. 26, no. 10, pp. 17013–17026, 2025.

[8] Kezhong Liu, Yiwen Zhou, Mozi Chen, Jianhua He, Jingao Xu, Zheng Yang, Chris Xiaoxuan Lu, and **Shengkai Zhang**. *VISC: mmWave Radar Scene Flow Estimation Using Pervasive Visual-Inertial Supervision*. IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2025.

[9] Cong Fan, **Shengkai Zhang**, Kezhong Liu, Shuai Wang, Zheng Yang, and Wei Wang. *Enhancing mmWave Radar Point Cloud via Visual-Inertial Supervision*. IEEE International Conference on Robotics and Automation (ICRA), 2024.

[10] Danyang Li, Yishujie Zhao, Jintao Xu, **Shengkai Zhang**, Longfei Shangguan, and Zheng Yang. *edgeSLAM2: Rethinking Edge-Assisted Visual SLAM with On-Chip Intelligence*. IEEE International Conference on Computer Communications (INFOCOM), pp. 1–10, 2024.

[11] Hongzhou Li, Sijie Yu, **Shengkai Zhang**, and Guang Tan. *Resolving Loop Closure Confusion in Repetitive Environments for Visual SLAM through AI Foundation Models Assistance*. IEEE International Conference on Robotics and Automation (ICRA), 2024.

[12] Zheng Li, Jun Ma, Feifeng Jiang, and **Shengkai Zhang**. *Assessing the Impacts of Urban Morphological Factors on Urban Building Energy Modeling Based on Spatial Proximity Analysis and Explainable Machine Learning.* Journal of Building Engineering, vol. 85, no. 7, article no. 108675, 2024.

[13] Dashuai Pei, Danei Gong, Kezhong Liu, Xuming Zeng, **Shengkai Zhang**, Mozi Chen, and Kai Zheng. *mmCTD: Concealed Threat Detection for Cruise Ships via mmWave Radar*. IEEE Transactions on Vehicular Technology, vol. 73, no. 12, pp. 18434–18451, 2024.

[14] Pengen Gao, **Shengkai Zhang**, Wei Wang, and Chris Xiaoxuan Lu. *Robust Metric Localization in Autonomous Driving via Doppler Compensation with Single-Chip Radar*. IEEE Transactions on Intelligent Transportation Systems, vol. 25, no. 1, pp. 491–502, 2024.

[15] Danei Gong, Kezhong Liu, Dashuai Pei, Haoran Zhang, **Shengkai Zhang**, and Mozi Chen. *Wi-Watch: Wi-Fi-Based Vigilant-Activity Recognition for Ship Bridge Watchkeeping Officers*. IEEE Transactions on Instrumentation and Measurement, vol. 73, article no. 9503717, 2023.

[16] Kezhong Liu, Dashuai Pei, **Shengkai Zhang**, Mozi Chen, Xuming Zeng, Kai Zheng, and Chunshen Li. *WiCrew: Gait-based Crew Identification for Cruise Ships Using Commodity WiFi*. IEEE Internet of Things Journal, vol. 10, no. 8, pp. 6960-6972, 2022.

[17] Fei Xiao, **Shengkai Zhang**, Sheyang Tang, Shaojie Shen, Huixin Dong, and Yi Zhong. *WiSion: Bolstering MAV 3D Indoor State Estimation by Embracing Multipath of WiFi*. IEEE Transactions on Vehicular Technology, vol. 72, no. 1, pp. 253-266, 2023.

[18] **Shengkai Zhang**, Wei Wang, Ning Zhang, Tao Jiang. *LoRa Backscatter Assisted State Estimator for Micro Aerial Vehicles with Online Initialization.* IEEE Transactions on Mobile Computing (TMC), vol. 21, no. 11, pp. 4038-4050, 2022.

[19] Pengen Gao, **Shengkai Zhang**, Wei Wang, and Chris Xiaoxuan Lu. *DC-Loc: Accurate Automotive Radar Based Metric Localization with Explicit Doppler Compensation.* IEEE International Conference on Robotics and Automation (ICRA), 2022.

[20] Kezhong Liu, Wen Yang, Mozi Chen, Kai Zheng, Xuming Zeng, **Shengkai Zhang**, and Cong Liu. *Deep Learning based Wireless Human Motion Tracking for Mobile Ship Environments.* IEEE Internet of Things Journal, vol. 9, no. 23, pp. 24186-24198, 2022.

[21] **Shengkai Zhang**, Sheyang Tang, Wei Wang, Tao Jiang, and Qian Zhang. *Conquering Textureless with RF-referenced Monocular Vision for MAV State Estimation.* IEEE International Conference on Robotics and Automation (ICRA), 2021.

[22] **Shengkai Zhang**, Wei Wang, Tao Jiang. *WiFi-Inertial Indoor Pose Estimation for Micro Aerial Vehicles.* IEEE Transactions on Industrial Electronics (TIE), vol. 68, no. 5, pp. 4331-4340, 2021.

[23] **Shengkai Zhang**, Wei Wang, Sheyang Tang, Tao Jiang. *Robot-assisted Backscatter Localization for IoT Applications.* IEEE Transactions on Wireless Communications (TWC), vol. 19, no. 9, pp. 5807-5818, Sep. 2020.

[24] **Shengkai Zhang**, Wei Wang, Ning Zhang, Tao Jiang. *RF Backscatter-based State Estimation for Micro Aerial Vehicles.* IEEE International Conference on Computer Communications (INFOCOM), 2020.

[25] **Shengkai Zhang**, Wei Wang, Sheyang Tang, Tao Jiang. *Localizing Backscatters by a Single Robot With Zero Start-up Cost.* IEEE Global Communications Conference (GLOBECOM), 2019.

[26] **Shengkai Zhang**, Sheyang Tang, Wei Wang, Tao Jiang. *WINS: WiFi-Inertial Indoor State Estimation for MAVs.* ACM Conference on Embedded Networked Sensor Systems (SenSys) (Demo Session), 2018.

[27] Chen Chen, Wei Wang, **Shengkai Zhang**, and Bo Li. *Cluster Fair Queueing: Speeding up Data-Parallel Jobs with Delay Guarantees.* IEEE International Conference on Computer Communications (INFOCOM), 2017.

[28] Bing Li, **Shengkai Zhang**, and Shaojie Shen. *CSI-Based WiFi-Inertial State Estimation.* IEEE International Conference on Multisensor Fusion and Integration for Intelligent Systems (MFI), 2016.

[29] **Shengkai Zhang**, Bo Li, and Baochun Li. *Presto: Towards Fair and Efficient HTTP Adaptive Streaming from Multiple Servers.* IEEE International Conference on Communications (ICC), 2015.

[30] Jingjie Jiang, **Shengkai Zhang**, Bo Li, and Baochun Li. *Maximized Cellular Traffic Offloading via Device-to-Device Content Sharing.* Device-to-Device Communications in Cellular Networks, IEEE Journal on Selected Areas in Communications (JSAC), vol. 34, no. 1, pp.82-91, 2015.

[31] **Shengkai Zhang**, Guang Tan, Hongbo Jiang, Bo Li, and Chonggang Wang. *On the Utility of Concave Nodes in Geometric Processing of Large-Scale Sensor Networks.* IEEE Transactions on Wireless Communications (TWC), vol. 13, no. 1, pp.132-143, 2014.

[32] Hongbo Jiang, **Shengkai Zhang**, Guang Tan, Chonggang Wang. *Connectivity-based Boundary Extraction of Large-Scale 3D Sensor Networks: Algorithm and Applications.* IEEE Transactions on Parallel and Distributed Systems (TPDS), 2014.

[33] Guang Tan, Hongbo Jiang, **Shengkai Zhang**, Zhimeng Yin, Anne-Marie Kemarrec. *Connectivity-based and Anchor-free Localization in Large-scale 2D/3D Sensor Networks.* ACM Transactions on Sensor Networks (TOSN), vol. 10, no. 1, Article 6 (November 2013), 2013.

[34] Hongbo Jiang, **Shengkai Zhang**, Guang Tan, Chonggang Wang. *CABET: Connectivity-based boundary extraction of large-scale 3D sensor networks.* IEEE International Conference on Computer Communications (INFOCOM), 2011.

[35] Guang Tan, Hongbo Jiang, **Shengkai Zhang**, and Anne-Marie Kermarrec. *Connectivity-based and Anchor-Free Localization in Large-Scale 2D/3D Sensor Networks.* ACM International Symposium on Mobile Ad Hoc Networking and Computing (MOBIHOC), 2010.

## PATENTS

[1] Mozi Chen, Yifan Zhang, Mengda Chen, Kezhong Liu, Kai Zheng, Xuming Zeng, and **Shengkai Zhang**. *Method and Device for Monitoring Ship-Operator Activity Based on Physical Channels*. China, ZL202310492596.0, May 5, 2023.

[2] Mozi Chen, Bowen Yin, Dashuai Pei, Kezhong Liu, Xing Yang, **Shengkai Zhang**, and Xuming Zeng. *Method for Assessing the Vigilance of Ship Watchkeeping Personnel Based on Wireless Signals*. China, ZL202310483640.1, May 4, 2023.

[3] Wei Wang, **Shengkai Zhang**, and Tao Jiang. *Intelligent Device Navigation Method and Navigation System*. United States, US011255677B2, February 22, 2022.

[4] **Shengkai Zhang**, Wei Wang, and Tao Jiang. *Intelligent Navigation Method, Apparatus, and Device*. China, ZL202011552550.6, August 2, 2022.

[5] Kezhong Liu, Dashuai Pei, Mozi Chen, **Shengkai Zhang**, Xuming Zeng, Chunshen Li, and Xun Zhang. *Velocity-Invariant Gait Recognition Method Based on WiFi Signals*. China, ZL202210659791.3, June 13, 2022.

[6] Wei Wang, **Shengkai Zhang**, and Tao Jiang. *Micro Aerial Vehicle Navigation Method Based on Passive RF Tags*. China, ZL201911042184.7, June 16, 2021.

[7] Wei Wang, **Shengkai Zhang**, Tao Jiang, and Sheyang Tang. *Robot-Assisted Passive Localization Method, System, and Device*. China, ZL201910492636.5, May 18, 2021.

[8] Pan Hui, **Shengkai Zhang**, and Christoph Peylo. *Method and System to Estimate User Desired Delay for Resource Allocation for Mobile-Cloud Applications*. Europe, EP2919438B1, June 19, 2019.

## RESEARCH FUNDING

- **National Natural Science Foundation of China (NSFC), General Program**, Grant No. 52671464, *Perception Methods for Unmanned Surface Vessels in Low-Visibility Environments Based on Vision Foundation Models*, 2027–2030. Principal Investigator.
- **National Natural Science Foundation of China (NSFC), General Program**, Grant No. 52571413, *Intelligent Collision-Avoidance Methods for Ships in Complex Scenarios Based on Large Language Models*, 2026–2029. Participant; ongoing.
- **National Natural Science Foundation of China (NSFC), General Program**, Grant No. 62171192, *Theory and Methods for Sub-6 GHz Wireless Channel Parameter Acquisition and Characteristic Analysis*, 2022–2025. Participant; completed.
- **Fundamental Research Funds for the Central Universities, Independent Innovation Project**, *State Estimation for Unmanned Aerial Vehicles Using Wireless and Visual Sensing*, 2021–2022. Principal Investigator; completed.

## HONOURS AND AWARDS

- **2025**: First Prize, Science and Technology Award of the China Institute of Navigation, for *Technologies and Applications for Personnel Safety Management and Man-Overboard Rescue on Large Ships.*
- **2025**: Second Prize, Hubei Provincial Science and Technology Progress Award, for *Key Technologies and Applications for Robust Localization and Tracking in Complex Ship-Port Environments.*
- **2024**: First Prize, Wuhan University of Technology Science and Technology Award, for *Key Technologies, Systems, and Industrial Applications for Robust Localization and Tracking in Complex Dynamic Interference Environments* (Certificate No. 2024-J-1-09).
- **2021**: Outstanding Paper Award from the Chinese Institute of Electronics for the IEEE Transactions on Wireless Communications paper *Robot-Assisted Backscatter Localization for IoT Applications.*
- **2020**: Huawei Scholarship, First Class.
- **2020**: National Scholarship for Doctoral Students (ranked first; two recipients in the school).
- **2020**: Outstanding Doctoral Academic Scholarship, First Class.
- **2019**: First Prize, 4th HUST Graduate Future Aircraft Innovation Competition (first place university-wide).
- **2019**: HUST Graduate Innovation and Entrepreneurship Fund (Grant No. 2019ygscxcy009).
- **2019**: HUST Zhixing Scholarship.
- **2018**: Third Prize, 7th HUST Qiushi Cup Extracurricular Academic Science and Technology Competition.
- **2018**: HUST Outstanding Graduate Student.
- **2017–2020**: HUST Doctoral Academic Scholarship.
- **2015**: Student Travel Grant, IEEE ICC, London.
- **2012–2014**: HKUST Postgraduate Scholarship.
- **2009–2012**: HUST Master's Academic Scholarship.

## TEACHING EXPERIENCE

- 2024 Spring: Indoor Localization Technology (WUT / School of Information Engineering)
- 2024 Spring: Digital Electronics (WUT / School of Information Engineering)
- 2022 Fall: Introduction to Information Technology (WUT / School of Information Engineering)